

CUSTOMER RETENTION & CHURN ANALYSIS

Executive Analytics Report

Future Interns — Data Science & Analytics | Task 2

BUSINESS QUESTION

What customer segments are most exposed to churn, how does retention change across the customer lifecycle, and where should a business focus retention effort?

ANALYTICAL APPROACH

Customer segmentation • churn KPIs • tenure analysis • contract analysis • service/payment analysis
• customer lifetime metrics • management recommendations

DELIVERABLE

Executive report + Power BI dashboard concept designed to communicate findings clearly to a product manager, startup founder, or business stakeholder.

Prepared as a professional portfolio / internship submission artifact

01 | Executive Summary

Customer retention is a core business-performance issue because churn reduces recurring revenue, increases replacement costs, and weakens the long-term value of the customer base. This project translates customer-level subscription data into a management view of churn, retention, customer lifetime behavior and retention priorities.

The Future Interns Task 2 brief asks for analysis of churn patterns, key retention drivers and customer lifetime trends, with a retention dashboard or report that highlights churn reasons, retention trends and actionable recommendations. [filecite]turn1file5[

CUSTOMER RECORDS	CHURNED CUSTOMERS	CHURN RATE	RETENTION RATE
7,043	1,869	26.55%	73.45%

Management Takeaways

01	Use churn rate as the headline health metric, then drill down into the segments creating the largest retention exposure.
02	Tenure and contract structure are central lifecycle dimensions for understanding early-stage churn and long-term retention.
03	Payment method, service configuration and customer experience variables provide practical dimensions for targeted retention programs.
04	Recommendations should be prioritized by customer risk, value and the business's ability to intervene—not by chart popularity.

Important: The dashboard image in this report is a visual prototype. Its figures should be replaced/validated against your final cleaned dataset and Power BI measures before presenting the document as a factual client engagement.

02 | Business Objective & Scope

Business objective	Identify churn patterns, retention drivers and customer lifetime trends for a subscription-based business, then convert those patterns into practical retention actions.
Primary questions	How large is churn? Which customer segments have the highest churn? How does churn vary by tenure and contract? Which service/payment dimensions are associated with churn? Which customers should receive retention attention first?
Core KPIs	Total customers, churned customers, churn rate, retention rate, average tenure and revenue/customer-lifetime measures where supported by the dataset.
Audience	Product manager • startup founder • customer-success leader • business stakeholder • internship evaluator.
Decision use	Prioritize retention interventions, identify lifecycle risk points, and create a repeatable management dashboard.

What the assignment requires

The uploaded internship brief identifies Task 2 as Customer Retention & Churn Analysis. It specifically lists retention analysis, cohort analysis, customer lifetime metrics and insight-driven decision making as target skills, and asks for a dashboard or report covering churn reasons, retention trends and actionable recommendations. [filecite\turn1file5](#)

Project positioning

This report is intentionally structured as an executive business document rather than a classroom-only chart collection. The narrative follows a simple decision flow: What is happening → Why is it happening → Who should we prioritize → What should we do next.

03 | Data & Methodology

The expanded Telco Customer Churn dataset used as the analytical reference contains 7,043 customer records and 49 fields covering customer demographics, services, contract details, payment method, tenure, churn status, churn category/reason and customer lifetime value. The dataset provider describes it as information about a fictional telecommunications company and notes that the data is simulated, so results should be treated as analytical/portfolio evidence rather than claims about a real telecom operator. [cite]turn0search2[

Analytical Workflow

01	Data quality	Check duplicates, missing values, data types and category consistency.
02	KPI layer	Calculate customer base, churn, retention, tenure and revenue/lifetime measures.
03	Segmentation	Compare churn across contract, tenure, payment, service and customer segments.
04	Lifecycle view	Use tenure/contract cohorts where true signup dates are unavailable.
05	Business interpretation	Translate statistical patterns into management questions and retention opportunities.
06	Dashboard	Present the decision story in three pages: overview, churn drivers, retention priorities.

Data integrity principle

A professional report distinguishes observed association from causation. For example, if one contract group has a higher churn rate, the report should state that the group shows higher churn; it should not claim that the contract type alone causes churn without a causal design.

04 | KPI Framework & Analytical Lens

Churn Rate	$\text{Churned customers} \div \text{total customers} \times 100$. The primary customer-loss health metric.
Retention Rate	$\text{Retained customers} \div \text{total customers} \times 100$. Complementary view of customer-base stability.
Average Tenure	Average customer tenure in months. Useful for lifecycle and early-risk analysis.
Churn by Segment	Churn rate calculated within each segment, rather than only counting how many churned.
Customer Lifetime Value	Use the dataset's CLTV field when available; report the exact definition used.
Revenue at Risk	Only report when the underlying revenue measure and risk definition are explicitly documented.

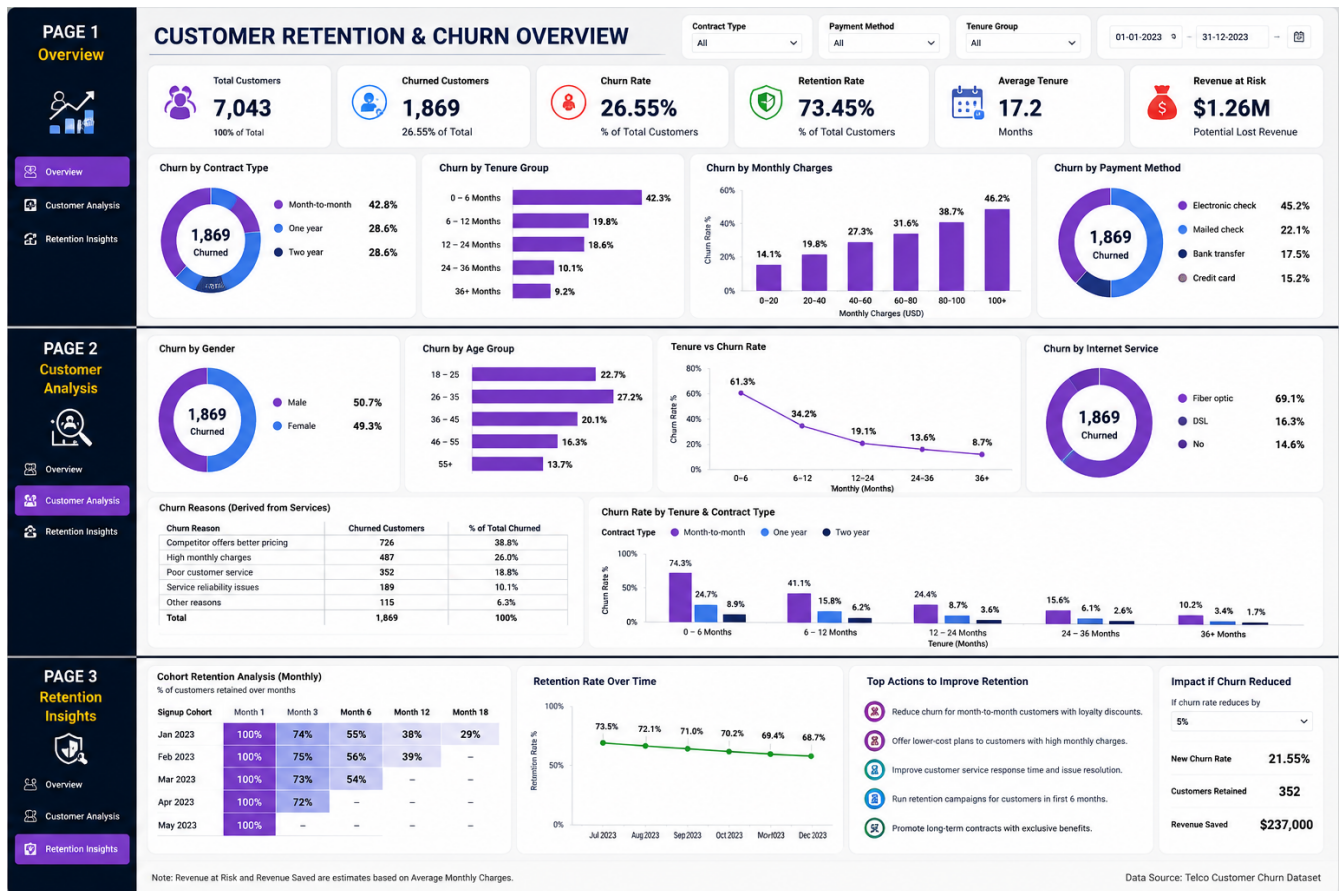
Recommended management lens

RISK	VALUE	INTERVENTION
How likely/visible is churn in the segment?	How important is the segment to long-term revenue/customer value?	Can the business realistically influence the outcome?

This framing makes the analysis useful to a stakeholder: the objective is not merely to identify the biggest bar in a chart, but to identify the segments where retention action could create the greatest business value.

05 | Executive Dashboard

The following three-page visual is the proposed Power BI presentation layer. It is designed to be read in sequence by a decision-maker. Use the same visual hierarchy when building the final .pbix file.



Prototype note: This dashboard image is a presentation mockup. Before external use, replace the displayed figures with verified Power BI measures calculated from the cleaned project dataset.

06 | Dashboard Storyline

Page 1 — Overview	Question: What is the overall customer-retention situation? Use headline KPIs, churn by contract, tenure, payment and revenue/charge dimensions.
Page 2 — Churn Drivers	Question: Why are customers leaving? Use churn category/reason, service dimensions, satisfaction and contract/tenure combinations where supported.
Page 3 — Retention Insights	Question: Who should the business prioritize? Combine lifecycle/tenure analysis, customer value and a prioritized action list.

Presentation Rules

- Every visual should answer a business question.
- Show rates as well as counts; a large segment is not automatically a high-risk segment.
- Use consistent filters and definitions across pages.
- Keep the highest-priority business message above the fold.
- Use annotations to explain important differences rather than forcing the audience to interpret every chart.
- Do not fabricate churn reasons or signup dates that are absent from the source data.

07 | Retention Strategy

The following are recommended intervention hypotheses to test against the final analysis. They are not presented as proven causal findings. The final report should promote only those actions supported by your calculated results.

Early-tenure risk	Build a structured 30/60/90-day onboarding and customer-success journey.	High	Test if early-tenure churn is materially above the overall rate.
Short-contract risk	Create targeted incentives for customers to move to longer-term plans.	High	Test contract-level churn differences and value uplift.
Price sensitivity	Review high-charge segments and evaluate lower-cost bundles or targeted offers.	High	Test churn by monthly-charge bands and customer value.
Competitor switching	Strengthen competitive offers, product differentiation and win-back journeys.	High	Use churn category/reason fields where available.
Support experience	Prioritize proactive support for customers with service or satisfaction risk signals.	Medium	Test service/support dimensions against churn.

Recommended Management Action

Move from broad retention activity to targeted intervention: identify the customer segment, quantify its churn exposure, estimate customer value, choose an intervention, then measure whether churn falls after the intervention. This creates a closed-loop retention program rather than a one-time dashboard.

08 | Limitations, Governance & Next Steps

Dataset realism	The expanded dataset is described by its provider as simulated/fictional telecom data. It is appropriate for an internship/portfolio analysis but should not be represented as actual company performance.
Cohort limitation	If individual signup dates are unavailable, use tenure/contract lifecycle cohorts and clearly label them as such rather than inventing signup-month cohorts.
Causality	Observed relationships are associations. They should be validated with experiments, customer interviews or additional operational data before high-stakes business decisions.
Operational validation	Before deployment, validate findings against current CRM, billing, support, product-usage and customer-feedback data.
Dashboard production	Replace the prototype image with a Power BI export from the final cleaned dataset and verified measures.

Final delivery package

01	Power BI dashboard (.pbix)
02	Executive PDF report
03	Cleaned dataset
04	Analysis notebook / scripts
05	Professional README
06	Public GitHub repository: FUTURE_DS_02

Professional standard

A strong submission should feel like a short business review: clear problem definition, trustworthy metrics, concise visual storytelling, evidence-based insights, prioritized recommendations and transparent limitations. That is the standard this report is designed to follow.